

Equivalence Completion Between Electroweak and Electromagnetic Coupling Bookkeeping

Amos Jay Maley

Abstract

We complete an equivalence map between electroweak gauge bookkeeping and low-energy electromagnetic bookkeeping. Parameters commonly treated as independent in the ultraviolet description are shown to overcount tensor load once admissible redescription and regime transport are enforced. The electromagnetic sector admits a single invariant interaction strength. The result is eliminative, invariant under admissible redescription, and introduces no new physics.

1 Apparent Independence in Electroweak Gauge Bookkeeping

In electroweak gauge bookkeeping, the interaction terms are written as

$$\mathcal{L}_{\text{gauge}} \supset g_2 W_\mu^a J^{a\mu} + g_1 B_\mu J_Y^\mu, \quad (1)$$

with the weak mixing angle defined by

$$\tan \theta_W = \frac{g_1}{g_2}. \quad (2)$$

The parameters g_2 , g_1 , and θ_W are often treated as independent physical inputs in this skin.

2 Admissible Electromagnetic Bookkeeping

An admissible low-energy description of electromagnetic interactions is given by quantum electrodynamics, in which all electromagnetic processes are governed by a single coupling e ,

$$\mathcal{L}_{\text{EM}} \supset e A_\mu J_{\text{EM}}^\mu. \quad (3)$$

No independent parameters corresponding to g_2 , g_1 , or θ_W appear in this skin.

3 Lemma 1: Cross-Skin Transport

Electroweak symmetry breaking induces a field rotation

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos \theta_W & \sin \theta_W \\ -\sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix}. \quad (4)$$

Under this admissible redescription, the electromagnetic coupling is identified as

$$e = g_2 \sin \theta_W = g_1 \cos \theta_W. \quad (5)$$

4 Lemma 2: Invariant Witness Counting

If a tensor load contains n independent degrees of freedom, admissible redescription and regime transport must preserve n independent invariant witnesses across all admissible skins describing the same regime. A skin exposing fewer invariant witnesses implies redundancy in higher-dimensional parameterizations.

5 Failure of Independence

Assume (g_2, g_1) constitute independent tensor degrees of freedom for electromagnetic interaction strength. By Lemma 2, two independent invariant witnesses must survive admissible transport. The electromagnetic skin exposes only one invariant witness, e . Preserving independence would require loss of tensor load, hidden unwitnessed structure, or representational privilege. Each constitutes standing failure.

6 Forced Tensor Relation

The electromagnetic tensor load is uniquely specified by

$$e = g_2 \sin \theta_W = g_1 \cos \theta_W, \quad \tan \theta_W = \frac{g_1}{g_2} \quad (6)$$

For the electromagnetic sector, g_2 , g_1 , and θ_W are not independently admissible tensor degrees once e is fixed.

7 Corollaries

- The electromagnetic sector admits a one-dimensional tensor load.
- Electroweak and electromagnetic descriptions are tensor-identical for EM interactions after redundancy elimination.
- Any framework asserting two independently tunable electromagnetic invariant couplings violates standing unless it introduces forbidden tensor load.

Conclusion

Equivalence completion enforces collapse of apparent freedom in electroweak coupling bookkeeping for the electromagnetic sector. The result is eliminative, representation-invariant, and irreversible under admissible redescription.